

4.4 NOTE: In the 1st printing of PRML, the text of the exercise refers equation (4.23) where it should refer to (4.22).

From (4.22) we can construct the Lagrangian function

$$L = \mathbf{w}^T(\mathbf{m}_2 - \mathbf{m}_1) + \lambda (\mathbf{w}^T \mathbf{w} - 1) .$$

Taking the gradient of L we obtain

$$\nabla L = \mathbf{m}_2 - \mathbf{m}_1 + 2\lambda \mathbf{w} \tag{100}$$

and setting this gradient to zero gives

$$\mathbf{w} = -\frac{1}{2\lambda}(\mathbf{m}_2 - \mathbf{m}_1)$$

from which it follows that $\mathbf{w} \propto \mathbf{m}_2 - \mathbf{m}_1$.

Week 8) Book exercise 4.5)

We note that

$$J(\underline{w}) = \frac{(\underline{m}_2 - \underline{m}_1)^2}{s_1^2 + s_2^2}$$

By observation

$$\stackrel{(4.22), (4.24) \text{ and } (4.20)}{=} \frac{\|\underline{w}^T(\underline{m}_2 - \underline{m}_1)\|^2}{\sum_{n \in C_1} (\underline{w}^T \underline{x}_n - \underline{m}_1)^2 + \sum_{n \in C_2} (\underline{w}^T \underline{x}_n - \underline{m}_2)^2} \quad (*)$$

Now if we define

$$S_B = (\underline{m}_2 - \underline{m}_1)(\underline{m}_2 - \underline{m}_1)^T$$

and

$$S_W = \underbrace{\sum_{n \in C_1} (\underline{x}_n - \underline{m}_1)(\underline{x}_n - \underline{m}_1)^T}_{S_{W,1}} + \underbrace{\sum_{n \in C_2} (\underline{x}_n - \underline{m}_2)(\underline{x}_n - \underline{m}_2)^T}_{S_{W,2}}$$

then the denominator on (*) equals

$$\begin{aligned} \|\underline{w}^T(\underline{m}_2 - \underline{m}_1)\|^2 &= (\underline{w}^T \underline{m}_2 - \underline{w}^T \underline{m}_1) (\underline{w}^T \underline{m}_2 - \underline{w}^T \underline{m}_1)^T \\ &= \underline{w}^T S_B \underline{w} \end{aligned}$$

and the numerator on (*) equals

$$\sum_{n \in \mathcal{L}_1} (\underline{w}^T \underline{x}_n - m_1)^2 + \sum_{n \in \mathcal{L}_2} (\underline{w}^T \underline{x}_n - m_2)^2$$

$$(4.23) \rightarrow = \sum_{n \in \mathcal{L}_1} (\underline{w}^T (\underline{x}_n - \underline{m}_1))^2 + \sum_{n \in \mathcal{L}_2} (\underline{w}^T (\underline{x}_n - \underline{m}_2))^2$$

$$= \underline{w}^T \underline{S}_{w,1} \underline{w} + \underline{w}^T \underline{S}_{w,2} \underline{w}$$

$$= \underline{w}^T \underline{S}_w \underline{w}$$

(we use $\underline{S}_{w,1}$ and $\underline{S}_{w,2}$ to denote the first and second part of $\underline{S}_w$).

It then follows that

$$J(\underline{w}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2} = \frac{\underline{w}^T \underline{S}_w \underline{w}}{\underline{w}^T \underline{S}_w \underline{w}}$$

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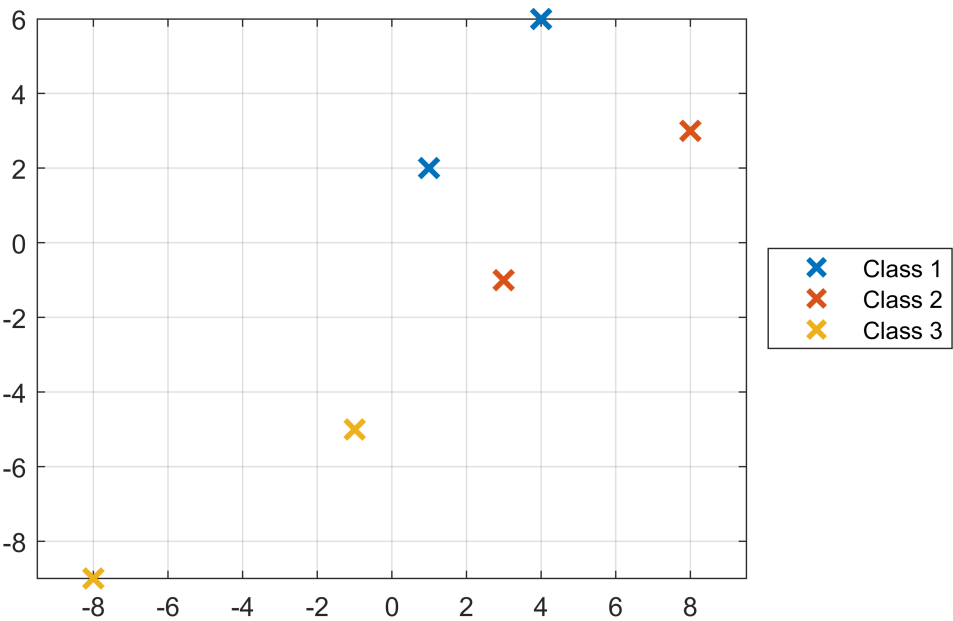
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Setup

```
X = 2x6
    1    4    8    3   -1   -8
    2    6    3   -1   -5   -9

l = 1x6
    1    1    2    2    3    3
```



1) Calculate the two-class projection vector w

Grap the correct data points and corresponding labels:

```
X1 = 2x2
    1    4
    2    6

X2 = 2x2
```

8	3
3	-1

Calculate the class means

m1 = 2×1
2.5000
4.0000

m2 = 2×1
5.5000
1.0000

Between-class covariance matrix S_B :

SB = 2×2
9 -9
-9 9

Within-class covariance matrix S_W :

SW1 = 2×2
4.5000 6.0000
6.0000 8.0000

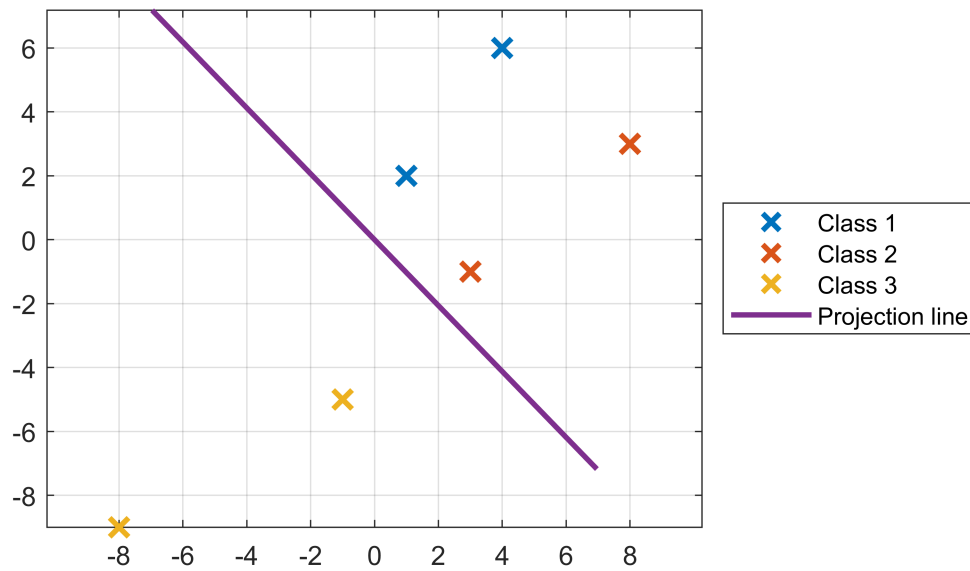
SW2 = 2×2
12.5000 10.0000
10.0000 8.0000

SW = 2×2
17 16
16 16

Projection/weight vector w:

w2 = 2×1
0.6961
-0.7179

2) Draw the projection vector w onto the scatter plot of the data points

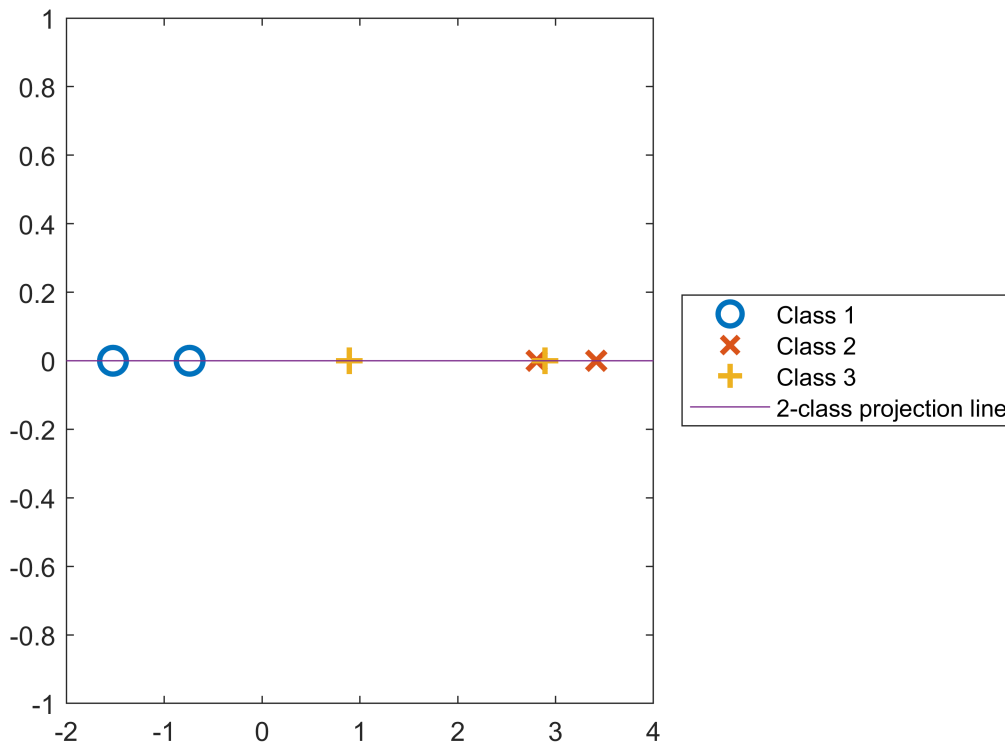


3) Can all data points be classified correctly using the two-class projection vector? Why/Why not?

Projecting X onto the projection vector w :

$$y = \frac{1}{\sqrt{6}} \begin{bmatrix} -0.7397 & -1.5228 & 3.4155 & 2.8063 & 2.8934 & 0.8919 \end{bmatrix}$$

$$1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 1 & 2 & 2 & 3 & 3 \end{bmatrix}$$



As seen from the figure, two of the data points from class 2 and class 3 are located almost ontop of eachother and with a small overlap. Therefore, class 2 and class 3 cannot be classified correctly using the FDA projection vector for class 1 and 2. However, class 1 is well separated from the two other classes, and can therefore be classified using the found projection vector.

4) Three-class projection vector w

```
m3 = 2x1
-4.5000
-7.0000
```

Within-class covariance matrix S_W :

```
SW3 = 2x2
24.5000  14.0000
14.0000   8.0000
```

```
SW = 2x2
41.5000  30.0000
30.0000  24.0000
```

Mean of all data points m :

```
m = 2x1
1.1667
-0.6667
```

```
m = 2x1
1.1667
```

-0.6667

"Between-class" covariance S_B :

SB1 = 2x2
3.5556 12.4444
12.4444 43.5556

SB2 = 2x2
37.5556 14.4444
14.4444 5.5556

SB3 = 2x2
64.2222 71.7778
71.7778 80.2222

SB = 2x2
105.3333 98.6667
98.6667 129.3333

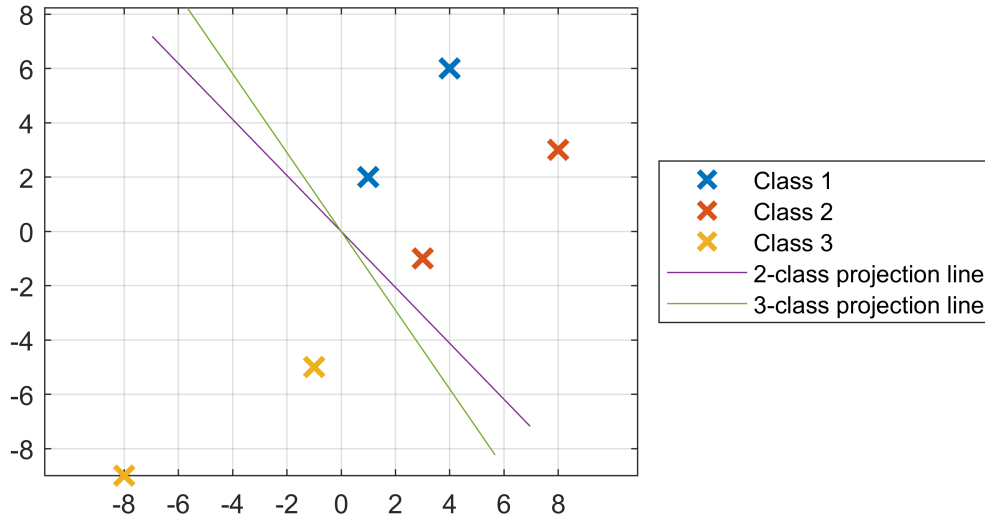
The projection vector can then be found as the eigenvector corresponding to the largest eigenvalue of $S_W^{-1}S_B$:

V = 2x2
-0.9201 0.5672
0.3917 -0.8236

D = 2x2
2.2044 0
0 18.3719

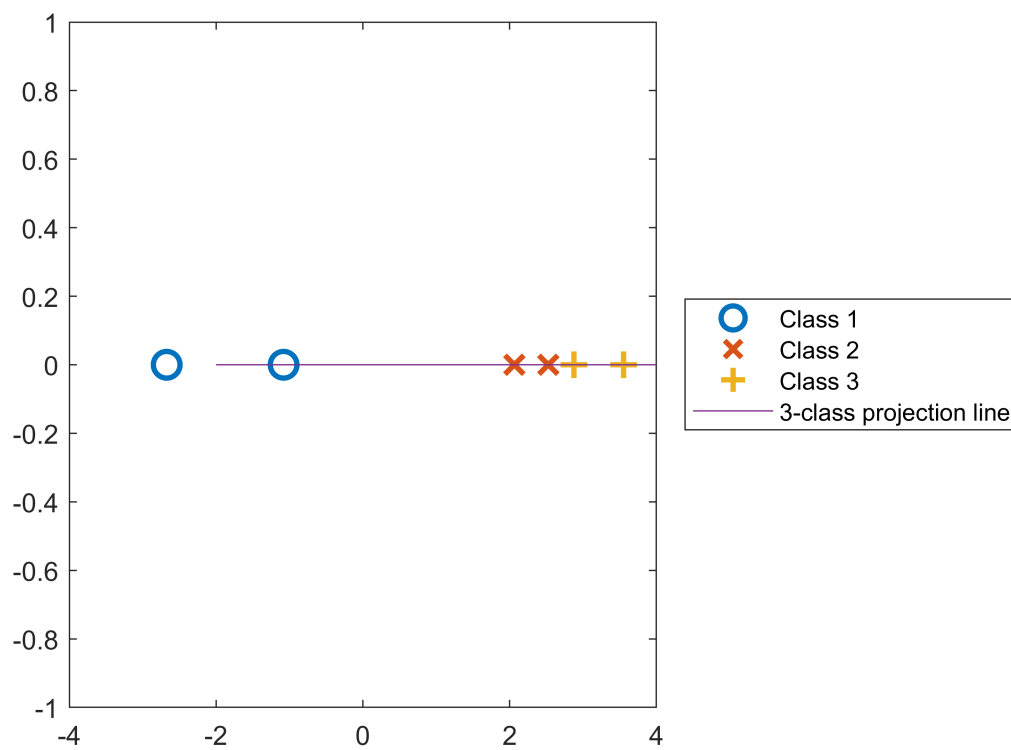
w3 = 2x1
0.5672
-0.8236

5) Draw the three-class projection vector w along with the data points



6) Can all data points be classified correctly using the three-class projection vector? Why/Why not?

y = 1×6
 -1.0801 -2.6731 2.0664 2.5251 3.5509 2.8753



As seen in the figure, there is no overlap between the three classes, hence they can all be classified correctly using two thresholds.